TRANVERSE AND LONGITUDINAL WAVES

QUESTION1

MULTIPLE CHOICE

Four possible options are provided as answers to the following questions. Each question has only ONE correct answer. Choose the answer and write only the letter (A – D) next to the question number (1.1 – 1.4) in the ANSWER BOOK,

for example 1.5 C.

1.1 The time taken for one complete wave cycle to pass a point is known as … (2)

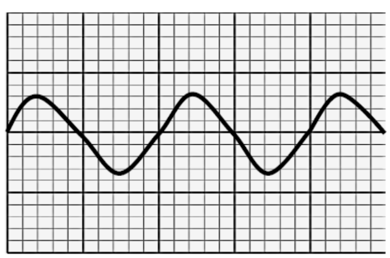
A Frequency.

B amplitude.

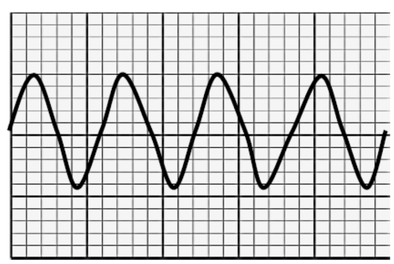
C Wavelength.

D Period.

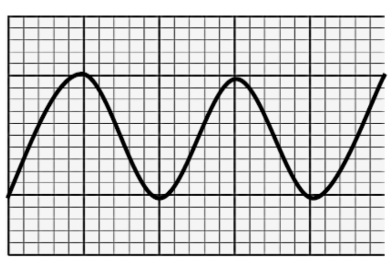
1.2 Study the following wave patterns: (2)



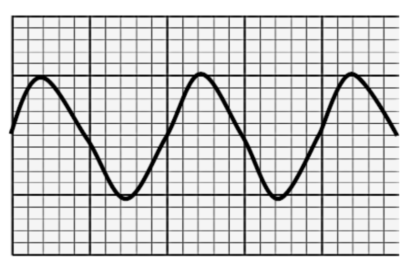
1



2



3



4

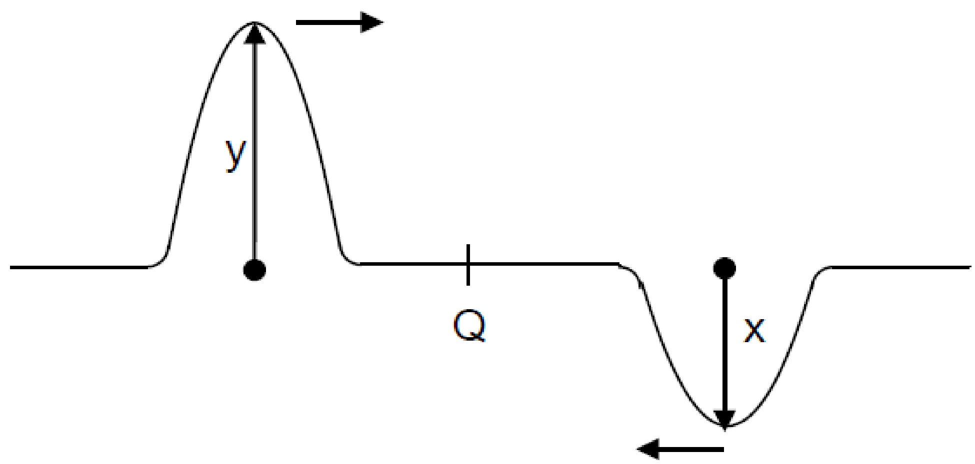
A 1 and 2

B 1 and 3

C 1 and 4

D 2 and 4

1.3 Two pulse are traveling towards each other along a string, as shown in the diagram below. (2)



A x+y

B 2(x+y)

C y-x

D 2(y-x)

1.4 A wave in which the particles of the medium vibrate at right angles to the path along which the wave travels through the medium, is produce by… (2)

A A bat

B A car's hooter

C An ambulance

D An X-ray machine

1.5 The number of waves passing a point every second is define as the…of the wave. (2)

A Speed

B Amplitude

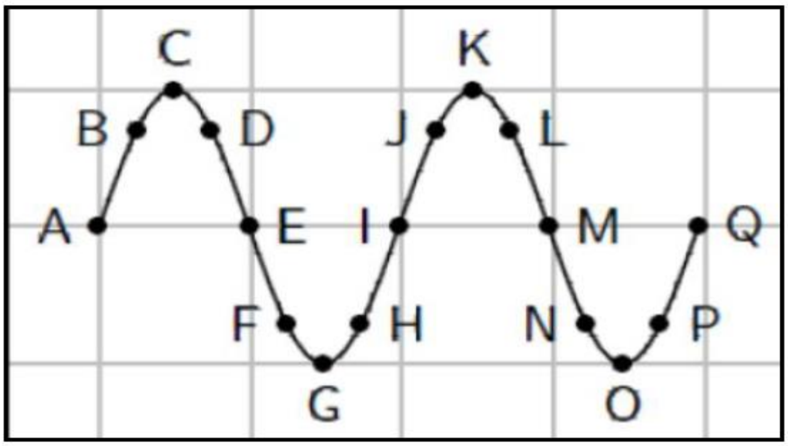
C Wavelength

D Frequency

LONG QUESTIONS

QUESTION 2

Study the following diagram and answer the questions:



Identify two sets of points that are in phase

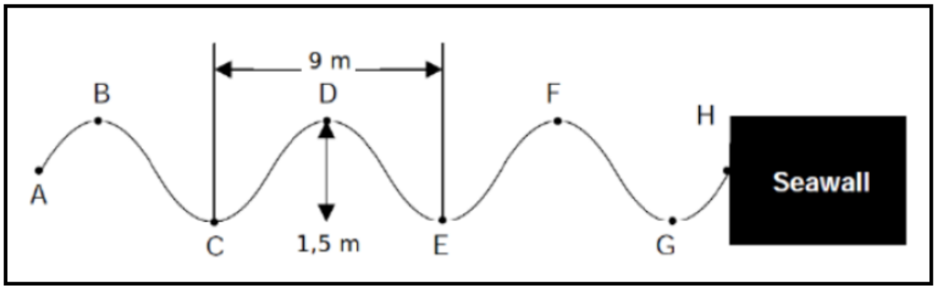
Identify two sets of points that are in phase.

Identify any two points that would indicate a wavelength

Identify any two points that would indicate a wavelength.

QUESTION 3

Water waves crash against a seawall around the harbour. Eight waves hit the seawall in 5s. The distance between successive troughs is 9 m. The height of the wave from trough to crest is 1,5 m.



3.1 How many complete waves are indicated in the sketch?

3.2 Write down the letters that indicate any TWO points that are:

3.2.1 In phase

3.2.2 Out of phase

3.2.3 Represent ONE wavelength

A. Calculate the amplitude of the wave.

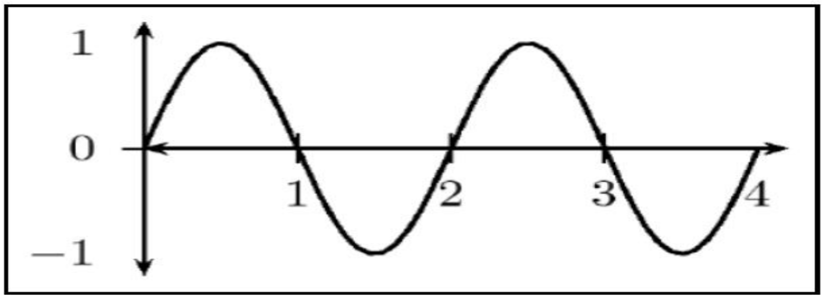
B. Show that the period of the wave is 0,67 s.

3.2.4 Calculate the frequency of the waves.

3.2.5 Calculate the velocity of the waves.

QUESTION 4

You are given the transverse wave below:



Draw the following:

4.1 A wave with TWICE the amplitude of the given wave.

4.2 A wave with HALF the amplitude of the given wave.

4.3 A wave travelling at the same speed with TWICE the frequency of the given wave.

4.4 A wave travelling at the same speed with HALF the frequency of the given wave

4.5 A wave with TWICE the wavelength of the given wave.

4.6 A wave with HALF the wavelength of the given wave

4.7 A wave travelling at the same speed with TWICE the period of the given wave.

4.8 A wave travelling at the same speed with HALF the period of the given wave.

QUESTION 5

A wave travels along a string at a speed of 1, 5m $\cdot$ s^{-1}. If the frequency of the source of the wave is 7,5 Hz, calculate:

5.1 The period of the wave. (3)

5.2 The wavelength of the wave. (4)

QUESTION 6

A heavy rope is flicked upwards, creating a single pulse in the rope.

Make a drawing of the rope and indicate the following in your drawing:

6.1 The direction of motion of the pulse

6.2 Amplitude

6.3 Pulse length

6.4 Position of rest

6.5 A pulse has a speed of 2, 5m · s^{-1}. How far will it have travelled in 6 s?

SOUND WAVES

QUESTION 7

Four options are provided as possible answers to the following questions. Each question has only ONE correct answer. Choose the answer and write only the letter (A–D) next to the question number (7.1–7.4) in the ANSWER BOOK, for example 1.3 C

7.1 Which ONE of the combinations below concerning the pitch and loudness of sound is CORRECT?

The pitch and loudness of sound depend on:

	PITCH	LOUDNESS
A	Frequency	Amplitude of vibration
B	Frequency	Speed of vibration
C	Amplitude of vibration	Frequency
D	Speed of vibration	Frequency

7.2 A wave in which the particles of the medium vibrate at right angles to the direction in which the wave travels through the medium, is produced by …

A Bat.

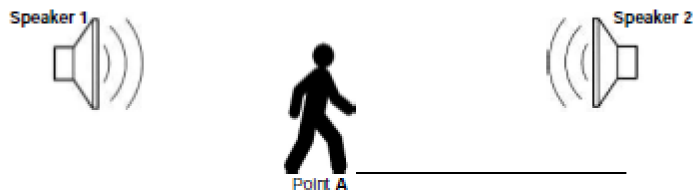
B A car's hooter

C An ambulance

D an X-ray machine

QUESTION 8

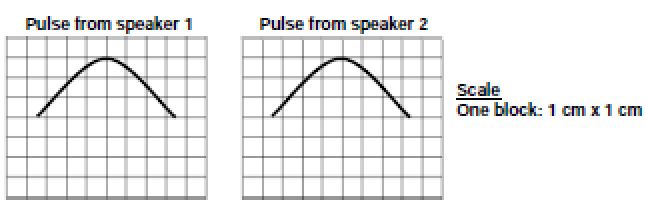
Learners are investigating the effect sound waves have on each other. They connect two speakers, which are several meters apart, to a signal generator and play the same signal with a single frequency through both speakers. One of the learners walks from speaker 1 to speaker 2, while listening carefully to the sound. At point **A** between the speakers, the learner observes that the sound is LOUDER than at any other point at either side of point **A**.



8.1 What type of wave is a SOUND wave?

8.2 Consider the following statement: *The addition of two pulses that occupy the same space at the same moment*.
Which PRINCIPLE is referred to here?

8.3 On an oscilloscope, pulses from speaker 1 and speaker 2 are as illustrated below.



8.3.1 Draw the wave pattern that will appear on the oscilloscope the moment the two pulses reach point **A** simultaneously. The drawing does not have to be to scale, but the magnitude of the amplitude of the pulse at **A** must be indicated on the drawing.

8.3.2 Identify the phenomenon in QUESTION 8.3.1.

9

9.1 Define the term *longitudinal wave.*

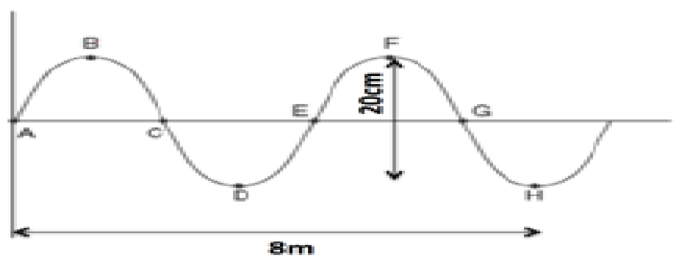
9.2 A sound wave travels to a high wall which is 225 m away from the source and is then reflected back.

If the speed of sound in air is 340 $m \cdot s^{-1}$, calculate the time it takes to hear the echo.

The same sound source used in QUESTION 2.2 above is used to produce an echo by sending the sound into water.

9.3 Is the time it takes to hear the echo LESS THAN, EQUAL TO or THE SAME as that obtained in QUESTION 9.2? Give a reason for the answer.

10 Study the wave pattern. It takes 0.3s to complete the pattern from A to H.



10.1 Define the term amplitude

10.2 Explain the difference between a period and frequency

10.3 Name any two points that are in phase.

10.4 Determine the wavelength of the wave

10.5 What is the amplitude of the wave?

10.6 What is the period of the wave?

10.7 Calculate the frequency of the wave

10.8 Calculate the speed of the wave

10.9 Determine how long it will take the wave to travel a distance of 20m

ELECTROMAGNETIC RADIATION

QUESTION 1

1.1 Consider the following statement concerning ultraviolet radiation :

(i) It cannot be reflected

(ii) It has a longer wavelength than gamma rays

(iii) It is radiated from the sun and the sun may be harmful to humans.

Which ONE of the following combinations is CORRECT?

A (i) and (ii) only

B (ii) and (iii) only

C (i) and (iii) only

D (i) , (ii) and (iii)

1.2 Red light of frequency f and a wavelength λ shines on an object. The red light is then replaced by light of higher energy. How do the frequency and the wavelength of light shining on the object now compare with that of red light?

	FREQUENCY	**WAVELENGTH**
A.	Greater than f	Remains the same(λ)
B.	Less than f	Greater than λ
C.	Greater than f	Less than λ
D.	Remains the same(f)	Less than λ

1.3 The frequency of a wave is defined as the…

A Lowest point on a wave.

B Time taken for one complete wave.

C Number of complete waves per second.

D Number of points in phase in a wavelength.

1.4 Which type of wave DOES NOT travel at $3x10^8 m.s^{-1}$?

A. Radio

B Gamma

C Sound

D Microwaves

QUESTION 2 (LONG QUESTIONS)

The table below shows an arrangement of electromagnetic radiation according to their frequencies.

TYPE OF RADIATION	TYPE OF FREQUENCY(Hz)
Radio waves	10^5 - 10^{10}
Micro waves	10^{10} -10^{11}
Infrared(IR)	10^{11} -10^{14}
Visible light	10^{14} -10^{15}
Ultra violet(UV)	10^{15} -10^{16}
X-rays	10^{16} -10^{18}
Gamma rays	10^{18} - 10^{21}

2.1 Write down TWO properties of electromagnetic waves.

2.2 Which radiation has the highest energy?

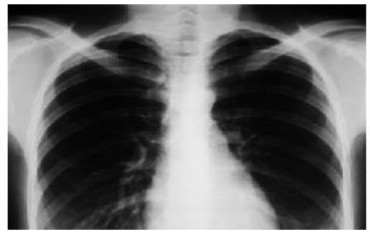
A certain radiation has energy of 1.99×10^{-20} J.

2.3 Identify the type of radiation associated with this energy.

2.4 Refer to diagrams **A** to **C** below.

A

B



C

Which type of radiation is used:

A

B

C

QUESTION 3

The types of electromagnetic radiation are arranged according to frequency in the table below

TYPE OF RADIATION	FREQUENCY(HZ)
Radio waves	10^5 - 10^{10}
Micro waves	10^{10} - 10^{11}
Infrared	10^{11} - 10^{14}
Visible light	10^{14} - 10^{15}
Ultraviolet	10^{15} - 10^{16}
X-rays	10^{16} - 10^{18}
Gamma rays	10^{18} - 10^{21}

3.1 How are electromagnetic waves generated?

3.2 What type of electromagnetic radiation has the highest energy?

3.3 Explain the answer to QUESTION 3.2

3.4 A certain type of electromagnetic radiation has a wavelength of 600×10^{-10}m

3.4.1 Identify the type of electromagnetic radiation by performing a Calculation

3.4.2 State ONE application of the type of radiation identified in QUESTION 3.4.1

QUESTION 4

The frequency and corresponding energy of electromagnetic waves are given in the table below.

WAVE	FREQUENCY(Hz)	ENERGY(J)
A	2×10^{9}	1.33×10^{-24}
B	4×10^{12}	2.65×10^{-21}
C	3.5×10^{15}	2.32×10^{-18}
D	1.8×10^{18}	1.19×10^{-15}
E	f	4.97×10^{-15}

4.1 Describe how an electromagnetic wave propagates

4.2 What is the relationship between frequency and energy of electromagnetic wave, as shown in the table above?

4.3 Calculate the:

4.3.1 Frequency of wave E

4.3.2 Wavelength of wave D

4.4 Which wave, A or B, has the HIGHER penetrative ability? Give a reason for the answer.